# AC Power Transmission

Power exchange, cables and OHL



confidential not for public disclosure

AMERICAN TERAWATT

# Abbreviations

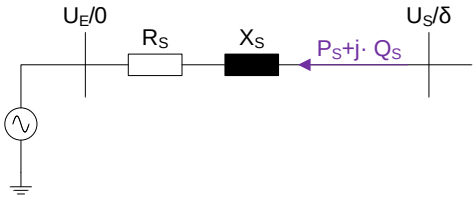
| | |
|---|---|
| AAAC | All-aluminium alloy conductor |
| AAC | All-aluminium conductor |
| AC | Alternating current |
| ACSR | Aluminium conductor steel reinforced |
| ACSS | Aluminium conductor steel supported |
| CB | Circuit breaker |
| DC | Direct current |
| DS | Disconnector (disconnect switch) |
| GMD | Geometric mean distance |
| GMR | Geometric mean radius |
| kcmil | Thousand circular mils |
| MOV | Metal-oxide varistor |
| OHL | Overhead line |
| PCC | Point of common coupling |
| RMS | Root mean square |
| SCL | Short-circuit level |
| SIL | Surge-impedance loading |
| SSCI | Sub-synchronous control interaction |
| SSR | Sub-synchronous resonance |
| VSC | Voltage-sourced converter |
| XLPE | Cross-linked polyethylene |
| pf | Power factor |
| p.u. | Per unit |

# Power transmission

Principle definition

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power exchange in the AC grid




$P_S$ is the export active power of the VSC converter station at PCC

$Q_S$ is the export reactive power of the VSC converter station at PCC

$U_S$ is the RMS value of the fundamental frequency component of the line-to-line AC voltage of the VSC converter station at PCC

$U_E$ is the equivalent AC voltage of the AC grid, corresponding to a SCL at PCC

$R_S$ is the equivalent resistance of the SCL impedance at PCC

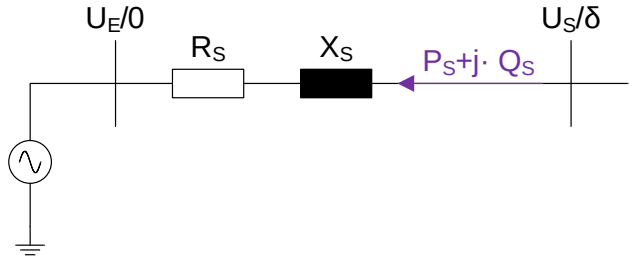
$X_S$ is the equivalent reactance of the SCL impedance at PCC

$\delta$ is the phase angle difference between AC voltages $U_S$ and $U_E$

IEC TR 63363-1:2022

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power exchange in the AC grid




$$P_S = \frac{U_S}{R_S^2+X_S^2} \cdot [(U_S - U_E \cdot \cos\delta) \cdot R_S + U_E \cdot X_S \cdot \sin\delta]$$

$$Q_S = \frac{U_S}{R_S^2+X_S^2} \cdot [(U_S - U_E \cdot \cos\delta) \cdot X_S - U_E \cdot R_S \cdot \sin\delta]$$

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

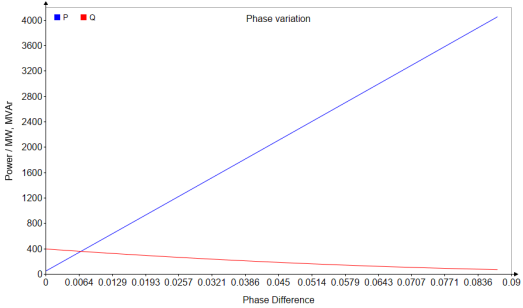
# Power exchange

| Parameter | Case 1 | Case 2 | Case 3 | Case 4 |
|---|---|---|---|---|
| $P_S$ | 1000 MW | 1500 MW | 1000 MW | $-1000$ MW |
| $U_E$ | 230 kV | 230 kV | 230 kV | 230 kV |
| $U_S$ | 232 kV | 232 kV | 240 kV | 232 kV |
| $R_S$ | 0.144 Ω | 0.144 Ω | 0.144 Ω | 0.144 Ω |
| $X_S$ | 1.15 Ω | 1.15 Ω | 1.15 Ω | 1.15 Ω |
| $\delta$ | 1.19° | 1.816° | 1.081° | $-1.319°$ |
| $Q_S$ | 288 MVAr | 239 MVAr | 905 MVAr | 541 MVAr |

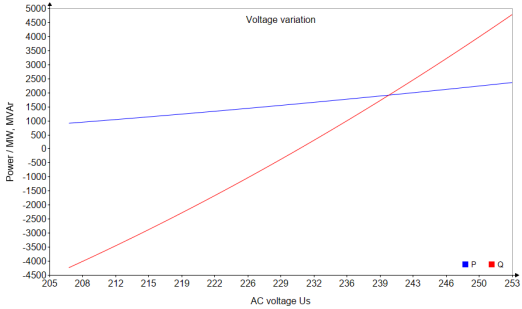
- Active power transmission requires reactive power.
- Higher AC voltage increases the reactive-power demand.
- Changing the power-flow direction also changes the reactive-power demand.


CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power exchange — active vs. reactive




Active power vs. voltage difference

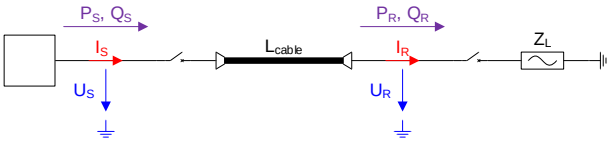



Active power vs. phase difference

Active power is transmitted via phase difference; reactive power via voltage amplitude.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power transmission in the AC grid



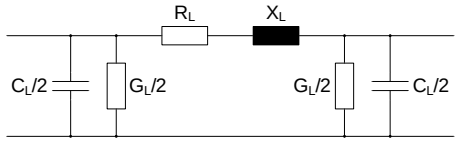

- AC power transmission principles shown with a simple AC network.
- The grid is connected to the load with impedance $Z_L$ via an AC cable of length $L_{cable}$.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Transmission line as 2-port units

| Param. | | OHL | Cable (XLPE) |
|---|---|---|---|
| $C'_L$ | nF/km | 10 | 180–420 |
| $R'_L$ | Ω/km | 0.1–1 | 0.1–1 |
| $X'_L$ | mH/km | 1 | 0.31–1.42 |
| $G'_L$ | µS/km | 0.05 | 1 |

- AC line represented by a $\pi$-section model.
- Cables are dominated by capacitance.
- OHL are dominated by inductance.

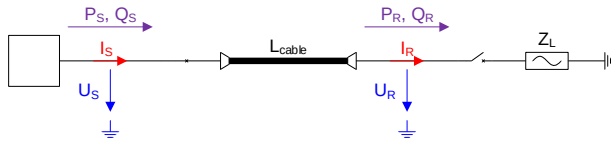



Example cable parameters:
$C'_L = 190\,\text{nF/km}$, $R'_L = 22\,\text{m}\Omega\text{/km}$,
$X'_L = 1.35\,\text{mH/km}$

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power transmission — no load connected

| Param. | Value |
|---|---|
| $U_{S,LL}$ | 230 kV (1.0 p.u.) |
| $I_S$ | 1060 A |
| $P_S$ | 2.2 MW |
| $Q_S$ | $-421$ MVAr |
| $U_{R,LL}$ | 1.22 p.u. |
| $I_R$ | 0 A |
| $L_{cable}$ | 60 mi |

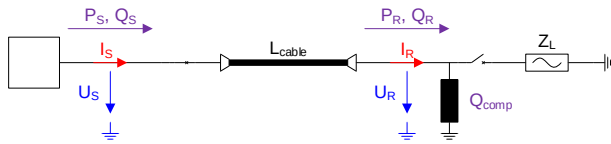



Connecting an AC cable creates significant current flow and reactive-power demand.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power transmission — reactive compensation

| Param. | Value |
|---|---|
| $U_{S,LL}$ | 230 kV (1.0 p.u.) |
| $I_S$ | 40 A |
| $P_S$ | 2.2 MW |
| $Q_S$ | $-14.4$ MVAr |
| $I_R$ | 830 A |
| $Q_R$ | $-277$ MVAr |
| $Q_{comp}$ | 400 MVAr |

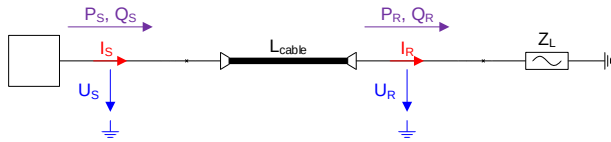



An uncompensated cable connection is usually not allowed; a shunt reactor $Q_{comp}$ is required.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power transmission — with load

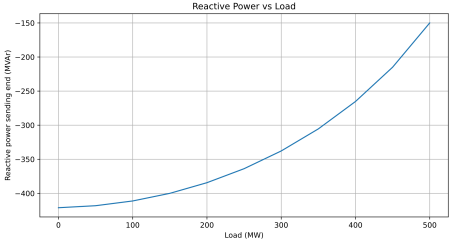
| Param. | Value |
|---|---|
| $U_{S,LL}$ | 230 kV (1.0 p.u.) |
| $I_S$ | 1340 A |
| $P_S$ | 511 MW |
| $Q_S$ | $-150$ MVAr |
| $U_{R,LL}$ | 1.06 p.u. |
| $I_R$ | 1180 A |
| $P_R$ | 500 MW |



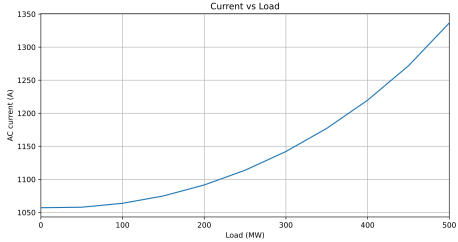

Power transmission reduces the voltage rise and the reactive-power demand.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power transmission — reactive power & current vs. active power



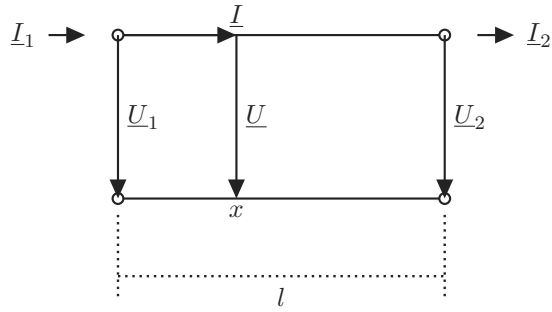

Reactive power vs. active power




AC current vs. active power

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# 2-port units

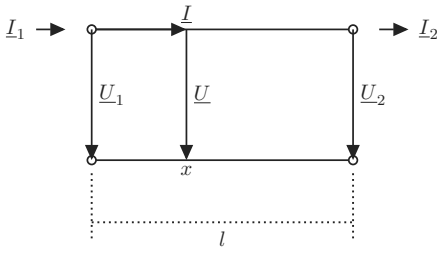



The calculation of the 2-port units is based on two paratemers which are defined by the physical set-up of the transmission:

- Propagation constant
- Wave impedance

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# 2-port units — line constants




Propagation constant

$$\underline{\gamma} = \alpha + j \cdot \beta = \sqrt{\underline{Z'} \cdot \underline{Y'}}$$

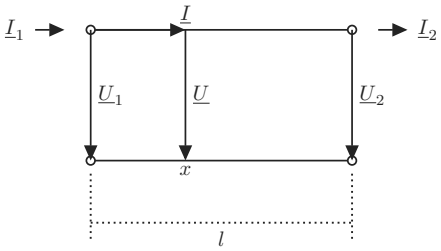
$$= \sqrt{(R'(\omega) + j \cdot \omega \cdot L'(\omega)) \cdot (G' + j \cdot \omega \cdot C')}$$

Wave impedance

$$\underline{Z}_w = \sqrt{\frac{\underline{Z'}}{\underline{Y'}}} = \sqrt{\frac{R'(\omega) + j \cdot \omega \cdot L'(\omega)}{G' + j \cdot \omega \cdot C'}}$$

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# 2-port units — line equations




Voltage along line, at distance $x$ from the sending end

$$\underline{U} = \underline{U}_2 \cosh \underline{\gamma}\,(l-x) + \underline{Z}_w\,\underline{I}_2 \sinh \underline{\gamma}\,(l-x)$$

$$\underline{I} = \frac{\underline{U}_2}{\underline{Z}_w} \sinh \underline{\gamma}\,(l-x) + \underline{I}_2 \cosh \underline{\gamma}\,(l-x)$$

2-port parameters, at the sending end $(x = 0)$

$$\underline{U}_1 = \underline{U}_2 \cosh \underline{\gamma} l + \underline{Z}_w\,\underline{I}_2 \sinh \underline{\gamma} l$$

$$\underline{I}_1 = \frac{\underline{U}_2}{\underline{Z}_w} \sinh \underline{\gamma} l + \underline{I}_2 \cosh \underline{\gamma} l$$

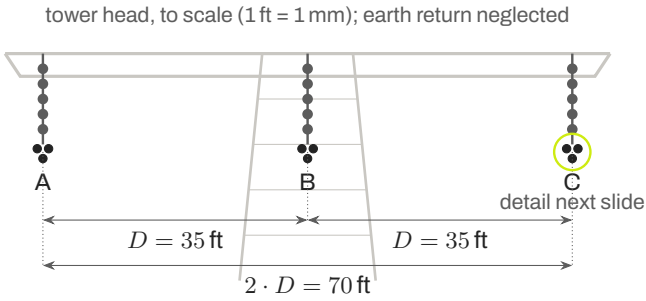
CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# OHL Configuration

Geometry of the 500 kV reference line

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Tower head — flat configuration




| Parameter | | Value |
|---|---|---|
| $D$ phase spacing | ft | 35 |
| $D_{AC}$ outer phases | ft | 70 |
| Conductors per phase | | 3 |
| $s$ bundle spacing | in | 18 |
| $d$ conductor diameter | in | 1.762 |
| $GMR$ conductor | ft | 0.0588 |

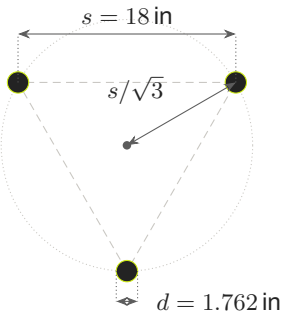
Bluebird ACSR, 2156 kcmil, 84/19, single circuit.

- Flat configuration: lowest towers, widest right-of-way; the outer phases see $2 \cdot D$.
- $GMD$ on the next slide assumes a transposed line.

Conductor data: Southwire ACSR bare conductor tables, catalog section 11-4. Tower body and insulator lengths are schematic.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Bundle detail and equivalent spacings




bundle, to scale (1 in = 1.6 mm)

$$GMR_b = \sqrt[3]{GMR \cdot s^2} \qquad \text{(bundle, } L'\text{)}$$

$$r_b = \sqrt[3]{\frac{d}{2} \cdot s^2} \qquad \text{(bundle, } C'\text{)}$$

$$GMD = \sqrt[3]{D \cdot D \cdot 2 \cdot D} = D \cdot \sqrt[3]{2} \qquad \text{(phases)}$$

$$L' = \frac{\mu_0}{2 \cdot \pi} \cdot \ln \frac{GMD}{GMR_b}, \qquad C' = \frac{2 \cdot \pi \cdot \varepsilon_0}{\ln (GMD / r_b)}$$

| **Quantity** | | $3\times$ | $2\times$ |
|---|---|---|---|
| $GMR_b$ | in | 6.11 | 3.56 |
| $r_b$ | in | 6.58 | 3.98 |
| $GMD$ | ft | 44.1 | 44.1 |
| $L'$ | mH/km | 0.892 | 1.000 |
| $C'$ | nF/km | 12.68 | 11.38 |
| $Z_w = \sqrt{L'/C'}$ | Ω | 265 | 296 |

For a two-conductor bundle $GMR_b = \sqrt{GMR \cdot s}$, $r_b = \sqrt{d/2 \cdot s}$. The third subconductor lowers $Z_w$ by $\approx 10$ % and so raises $P_{nat}$ by $\approx 11$ %.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power Transmission Limits

Thermal, voltage and stability limits of an OHL

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Three limits on the power transfer of a line

- **Thermal.** Conductor temperature sets the current; independent of length. Binding only on short lines.
- **Voltage drop.** The receiving-end voltage must stay in its band under load: $U_R \geq 0.95$ p.u. with $U_S = 1.0$ p.u.
- **Steady-state stability.** $\delta$ is kept well below $90°$ so the system survives the loss of a parallel element; $\delta \leq 30°$ leaves a factor 2 on the lossless limit.
- The *loadability* is the minimum of the three; against length it is the St. Clair curve.

$$P_{th} = \sqrt{3} \cdot U_n \cdot I_{th}$$

$$U_S = |A \cdot \underline{U}_R + B \cdot \underline{I}_R| \leq 1.0\,\text{p.u.}$$

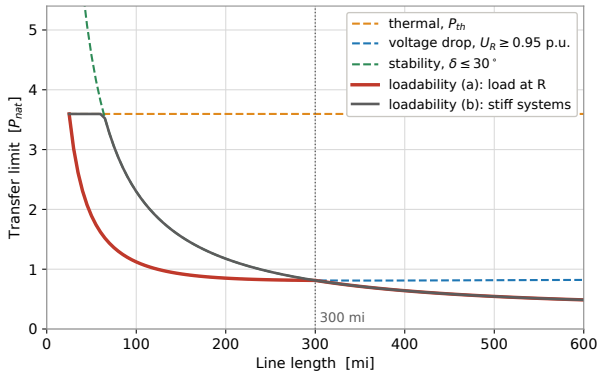
$$P_{stab} \approx \frac{U^2}{(1-k) \cdot X_L} \cdot \sin \delta_{max}$$

| Reference line | | Value |
|---|---|---|
| $I_{th}$, 3× Bluebird at 75 °C | A | 3900 |
| $P_{th}$ | MW | 3377 |
| $P_{nat}$ | MW | 939 |

$A$, $B$ two-port constants of the (compensated) line. The curves use the exact two-port power–angle relation for $P_{stab}$; the lossless form is shown for orientation.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Loadability of the 500 kV reference line

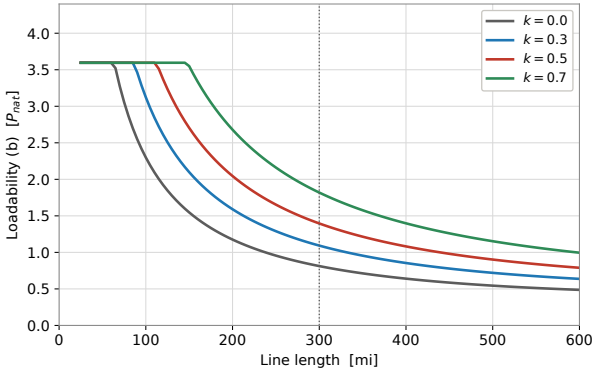



- **(a)** load at the receiving end, no reactive support: thermal to 25 mi, then voltage drop; falls through $1 \cdot P_{nat}$ at 125 mi.
- **(b)** two stiff systems, both ends at 1.0 p.u.: thermal to 60 mi, then angular stability; falls through $1 \cdot P_{nat}$ at 240 mi.
- At 300 mi both cases give 763 MW $(0.81 \cdot P_{nat})$ — the voltage and the angle criterion happen to coincide; the line cannot carry its natural power without compensation.

Uncompensated line. Voltage criterion with a unity power-factor load; a lagging load lowers it further, reactive support at the receiving end raises it.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Loadability with series compensation




| $k$ | $P_{stab}$ MW | $P_U$ MW | (b) $P_{nat}$ | thermal to mi |
|---|---|---|---|---|
| 0 | 763 | 763 | 0.81 | 60 |
| 0.3 | 1027 | 826 | 1.09 | 85 |
| 0.5 | 1311 | 844 | 1.40 | 110 |
| 0.7 | 1708 | 796 | 1.82 | 145 |

- Case (b): the bank acts on the stability limit — $k = 0.5$ lifts the 300 mi loadability from 0.81 to $1.40 \cdot P_{nat}$.
- $P_U$ of case (a) barely moves: the $I \cdot X_C$ rise is in quadrature with a unity-pf load, so reactive support at the receiving end is needed as well.

Values at 300 mi, midpoint bank. Curves show case (b), the planning case for a line between two strong systems.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Series Compensation

Shortening the electrical length of an OHL

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Series compensation — principle

A capacitor bank in series with the line cancels part of the series inductive reactance. The *degree of compensation* $k$ is defined against the total series reactance of the line:

$$k = \frac{X_C}{X_L}, \qquad X_L = \omega \cdot L' \cdot l, \qquad X_{eff} = (1-k) \cdot X_L$$

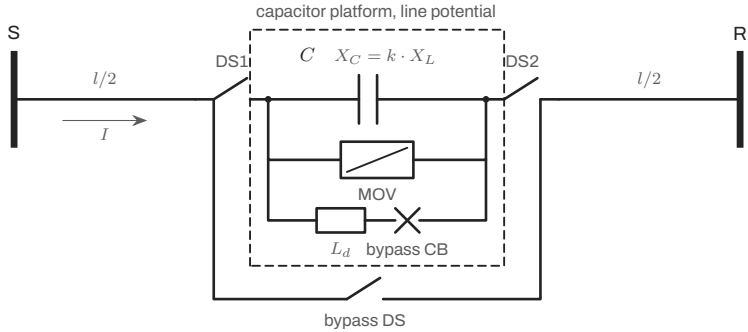
$$P = \frac{U^2}{(1-k) \cdot X_L} \cdot \sin\delta \quad \Rightarrow \quad P_{max} \approx \frac{U^2}{(1-k) \cdot X_L} \qquad \text{(lossless, } |U_S| = |U_R| = U\text{)}$$

- The line behaves as if it were shorter: for a given power the transmission angle $\delta$ falls and the steady-state stability margin rises.
- The rise $I \cdot X_C$ across the bank scales with the current: self-regulating, unlike shunt compensation.
- Practical range $k \approx 0.25 \ldots 0.70$; above that SSR, distance-protection behaviour and bank cost dominate.

$X_C$ reactance of the bank at $\omega$; $l$ line length; $\delta$ angle between sending- and receiving-end voltages.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Series compensation — single-line diagram




- MOV limits the capacitor voltage under fault current; the bypass CB with damping reactor $L_d$ takes over at the MOV energy limit.
- DS1/DS2 isolate the platform; the bypass DS keeps the line in service with the bank out ($k = 0$).

One phase shown; one platform per phase at the line midpoint of the reference case. Symbols IEC 60617.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Series compensation — reference line

The two-port model of the natural-power slides is reused; the bank is cascaded at the midpoint as a series element $\begin{bmatrix} 1 & -j \cdot X_C \\ 0 & 1 \end{bmatrix}$ between the two half-lines.

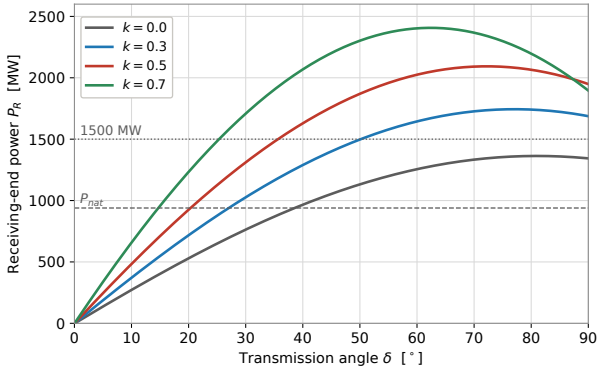
- Line constants from the conductor and tower geometry: $L' = \frac{\mu_0}{2 \cdot \pi} \cdot \ln \frac{D_m}{r_b}$, $C' = \frac{2 \cdot \pi \cdot \varepsilon_0}{\ln(D_m / r_{b_c})}$ with the bundle equivalent radii.
- $\beta \cdot l = 35.2°$: at 300 mi the line is well beyond the thermal plateau of the loadability curve.
- The compensation degree is swept over $k = 0,\ 0.3,\ 0.5,\ 0.7$.

| Quantity | | Value |
|---|---|---|
| $U_n$ | kV | 500 |
| $f$ | Hz | 60 |
| $l$ | mi (km) | 300 (483) |
| Conductor | | $3\times$ Bluebird ACSR |
| $R'$ (75 °C) | mΩ/km | 56.9 |
| $X'$ | Ω/km | 0.336 |
| $C'$ | nF/km | 12.7 |
| $X_L$ | Ω | 162.4 |
| $R$ | Ω | 27.5 |
| $Z_w$ | Ω | 266 |
| $P_{nat}$ | MW | 939 |

Flat configuration, 35 ft phase spacing, 18 in bundle spacing. Conductor data: Southwire ACSR tables, section 11-4.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Power–angle curves



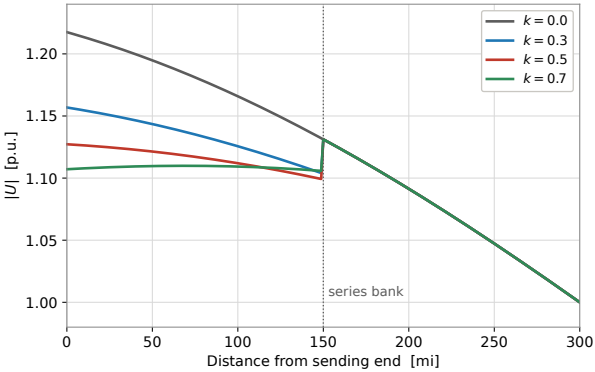

| $k$ | $X_C$ Ω | $P_{max}$ MW | $\delta$ at 1500 MW ° |
|---|---|---|---|
| 0 | 0 | 1363 | — |
| 0.3 | 48.7 | 1744 | 50.1 |
| 0.5 | 81.2 | 2093 | 35.7 |
| 0.7 | 113.7 | 2407 | 25.3 |

- Uncompensated, 1500 MW ($1.6 \cdot P_{nat}$) cannot be carried at all; $k = 0.5$ carries it at $36°$.
- $P_{max}$ moves to smaller angles as $k$ grows — the line gets electrically shorter.

Both terminals at 1.0 p.u.; exact two-port with losses and charging. The lossless $U^2/((1-k)\cdot X_L)$ would give 1540 MW uncompensated and 3080 MW at $k = 0.5$ — $R = 27.5\,\Omega$ and the charging are not negligible at this length.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Voltage profile at heavy load




- 900 MW at cos $\varphi = 0.95$ lagging delivered at the receiving end, $U_R$ held at 1.0 p.u.
- Uncompensated the sending end must sit at 1.217 p.u. — far outside the operating band.
- The midpoint bank lifts the voltage by $I \cdot X_C$ in the direction of power flow: $U_S$ falls to 1.157, 1.127 and 1.107 p.u. for $k = 0.3$, 0.5 and 0.7.
- The receiving half of the line is unchanged — the bank acts only on what is upstream of it, which is why end-of-line banks and split banks are also used.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Bank rating and protection

The bank is rated for the planned continuous line current, not for the conductor thermal limit:

$$U_C = I_{rated} \cdot X_C, \qquad Q_C = 3 \cdot I_{rated}^2 \cdot X_C$$

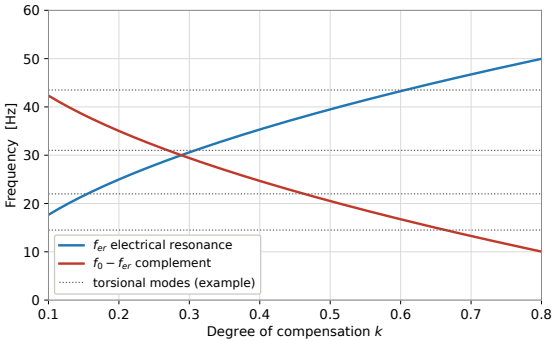
- Fault-current voltage is limited by a metal-oxide varistor (MOV) across the bank, protective level $2.0 \ldots 2.5 \cdot U_C$, with bypass breaker and damping circuit.
- Platform insulation at line potential; the bank carries the full fault current until the bypass closes.
- Emergency ratings (30 min, 10 s) are specified separately from the continuous rating.

| Quantity | | Value |
|---|---|---|
| $k$ | | 0.5 |
| $X_C$ | Ω | 81.2 |
| $C$ per phase | µF | 32.7 |
| $I_{rated}$ | A | 2000 |
| $U_C$ per phase | kV | 162 |
| $Q_C$ three-phase | Mvar | 974 |
| MOV level ($2.3 \cdot U_C$) | kV | 373 |

$I_{rated}$ chosen as $\approx 1.15\times$ the current at 1500 MW, 500 kV. Midpoint bank on the reference line.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE

# Sub-synchronous resonance — screening




Torsional modes shown are illustrative; real values come from the generator manufacturer.

Series resonance of bank, line and source:

$$f_{er} = f_0 \cdot \sqrt{\frac{X_C}{X_L + X_S}}$$

- Rotor currents see the complement $f_0 - f_{er}$, which must stay clear of the turbine-generator torsional modes (10–45 Hz for large steam units).
- $X_S = 25\,\Omega$ (15 % on 1500 MVA): $k = 0.5$ gives $f_{er} = 39.5$ Hz, complement 20.5 Hz — within 2 Hz of the 22 Hz example mode.
- A hit at screening level calls for a detailed SSR study or a different $k$; type-3 wind adds SSCI.

CONFIDENTIAL NOT FOR PUBLIC DISCLOSURE